nereus_Pi_shield_ADIN2111 — BOM for review

Board experiments/nick_raw_routing, layout freeze 2026-11-11 (tag s9-layout-freeze) · from `docs/design-review/bom_nick_raw_routing.csv` · 2026-11-11 · 53 lines, 108 fitted parts per board (+ 1 DNP) · 8 lines where the planned part differs from Sofar's (BOM for the JLC order of 20 boards; reasons and open lines: docs/design-review/bom_review_2026-11.md)

Refs	Qty	Sofar's part (specified)	Planned part	Why (planned note)	Backups (Alt1 / Alt2)	Sourcing	Footprint
C1, C3, C9-C15, C28, C53	11	**CL05A104KA5NNNC** Samsung	**CL05A104KA5NNNC** Samsung LCSC C100072	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-CL278-IPC_C
C2, C4	2	**CL05A105KP5NNNC** Samsung	**CL05A105KP5NNNC** Samsung LCSC C14445	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-CL511-IPC_C
C5, C6	2	**CC0402KRX5R6BB684** YAGEO	**CC0402KRX5R6BB684** Yageo LCSC C113784	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-CC0402-MFG
C7, C8	2	**CC0402JRNPO9BN150** Yageo	**CC0402JRNPO9BN150** Yageo LCSC C106997	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-CC0402-0_55-IPC_B
C16, C18, C24, C25	4	**08051C474KAT2A** KYOCERA AVX	**08051C474KAT2A** KYOCERA AVX LCSC C597304	As specified: Sofar's part is in JLC stock now (8,865 on 2026-11-11, need 82; JLC's listing shows no maker name). Planned before: YAGEO CC0805KKX7R0BB474, now ALT1	CC0805KKX7R0BB474 (YAGEO, C596323) 08051C474K4Z2A (KYOCERA AVX, C597303) *470 nF ±10 % 100 V X7R 0805; PoDL front end: Sofar review*	JLC stock (read 2026-11-11)	FP-0805-L_2_01_0_2-W_1_25-MFG
C17, C26	2	**C2012X7S2A105K125AB** TDK	**C2012X7S2A105KT000N** TDK LCSC C342785	Specified C2012X7S2A105K125AB has JLC stock 0 (2026-11-11) and Sofar's mote file gives no alternate. Same maker and series, 1 µF ±10 % 100 V X7S 0805, JLC stock 7,130 (need 42); thickness not confirmed (K125AB is 1.25 mm, KT000N doesn't state it)	GCM21BC72A105KE36L (Murata, C126585) *1 µF ±10 % 100 V X7S 0805; power path: Sofar review*	JLC stock (read 2026-11-11); not Sofar's part, see PLANNED NOTE	FP-C2012-125-0_2-MFG
C19, C27	2	**C0805C102MDRACTU** Kemet	**C0805C102MDRACTU** KEMET LCSC C2181610	As specified: Sofar's part is in JLC stock now (188 on 2026-11-11, need 40). Planned before: KEMET C0805C102KDRACTU, now ALT1	C0805C102KDRACTU (KEMET, C2167549) 0805B102K102CT (Walsin, C303890) *1 nF 1 kV X7R 0805, ±10 % (tighter); PoDL front end: Sofar review*	JLC stock (read 2026-11-11)	CAPC2013X145X50LL20T25
C20, C29, C49, C50, C54	5	**GRM188R72A104KA35D** Murata	**GRM188R72A104KA35D** Murata LCSC C77058	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-GRM188-0_1-e0_2_0_5-IPC_B
C21	1	**CL32B106KBJNNNE** Samsung	**CL32B106KBJNNNE** Samsung LCSC C138687	As specified: Sofar's part is in JLC stock now (77,751 on 2026-11-11, need 20). Planned before: Taiyo Yuden UMK325AB7106KM-T, now ALT2	1210B106K500NT (FH, C116808) UMK325AB7106KM-T (Taiyo Yuden, C386167) *10 µF ±10 % 50 V X7R 1210; power path: Sofar review*	JLC stock (read 2026-11-11)	FP-CL32-IPC_C
C22, C23	2	**EEEFT1H470AP** Panasonic	**EEEFT1H470AP** Panasonic LCSC C92059	As specified (Sofar's part, kept under SPEC constraint 8); Sofar's mote file gives no alternate. JLC public stock 0 on 2026-11-11, so from the parts bought		Nick's JLC parts library: order DOB202610210529840 (2026-10-20), qty 60, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	FP-FTD-MFG
C30, C55	2	**UMK107BBJ225KA-T** Taiyo Yuden	**UMK107BBJ225KA-T** Taiyo Yuden LCSC C268016	As specified; in JLC stock (read 2026-11-11); Taiyo Yuden now lists it as MSASU168BB5225KTNA01		JLC stock (read 2026-11-11)	FP-0603-L_1_6_0_2-W_0_8_0-MFG

C31	1	**C1608X5R0J226M080AC** TDK	**CL10A226MQ8NRNC** Samsung LCSC C59461	Specified C1608X5R0J226M080AC has JLC stock 1 (2026-11-11) and Sofar's mote file gives no alternate. Same spec, 22 µF ±20 % 6.3 V X5R 0603; JLC Basic part, stock 6,741,783 (need 30)	GRM188R60J226MEA0D (Murata, C77042) *22 µF ±20 % 6.3 V X5R 0603*	JLC stock (read 2026-11-11); not Sofar's part, see PLANNED NOTE	CAPC1608X100X20ML10
C32	1	**GRJ155R60J106ME11D** Murata	**GRJ155R60J106ME11D** Murata LCSC C469527	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-GRJ155-0_2-MFG
C33	1	**GRM155R60J475ME87D** Murata	**GRM155R60J475ME87D** Murata LCSC C76996	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-GRM155-0_15-MFG
C56, C57	2	**GRM21BR61C226ME44L** Murata	**GRM21BR61C226ME44L** Murata LCSC C86817	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	C_0805_2012Metric
C58	1	**GRM155R61A475MEAAD** Murata	**GRM155R61A475MEAAD** Murata LCSC C335105	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	C_0402_1005Metric
C59	1	**CL05A224KA5NNNC** Samsung	**CL05A224KA5NNNC** Samsung LCSC C47129	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-CL278-IPC_C
D1	1	**SMA6F33A** STMicroelectronics	**SMA6F33A-M3/H** Vishay LCSC C7083040	Sofar's own listed alternate 1 (mote file, MANUFACTURERPARTNUMBER1); the specified ST SMA6F33A is obsolete (JLC stock 2 on 2026-11-11)		Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 27, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	SMA6F33A-Footprint-1
D2, D3	2	**RB058LAM-60TFTR** ROHM	**RB058LAM-60TFTR** ROHM LCSC C3757753	As specified; JLC public stock 10 on 2026-11-11 (need 40), so from the parts bought		Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 45, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	FP-PMDTM-MFG
D4, D5	2	**CDSOD323-T36SC** Bourns Inc.	**CDSOD323-T36SC** Bourns LCSC C911382	As specified; in JLC stock (read 2026-11-11): 1,123, need 40. Pick the BOURNS listing; JLC has two other makers' parts of the same name		JLC stock (read 2026-11-11)	FP-SOD-323-MFG
D8	1	**KT-0603R** Hubei KENTO Elec	**KT-0603R** Hubei KENTO LCSC C2286	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	LED_0603_1608Metric
D9, D10	2	**KT-0603YG** Hubei KENTO Elec	**KT-0603YG** Hubei KENTO LCSC C2289	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	LED_0603_1608Metric
F1	1	**Polyfuse**	**BSMD1206-150-16V** BHFUSE LCSC C883133	1.5 A hold / 3 A trip; LiFePO4 charge bridge open (0.57 A); in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11). A PPTC resettable fuse: kept for this unpotted build on Nick's word (2026-11-11), an exception to SPEC constraint 8; replace it before a potted build	Fuse_1206_3216Metric
J1	1	**Raspberry_Pi_2_3**	**2.54-2*20** ZHOURI LCSC C2977589	2×20 2.54 mm female through-hole socket, 8.5 mm tall: Nick's decision of 2026-11-11 (the 8 or 8.5 mm header). JLC's attributes: insulation height 8.5 mm, pin length 3.2 mm, gold on brass, 2.5 A; JLC stock 37,633 (need 20). Datasheet not read	ZX-PM2.54-2-20PY (Megastar, C7499354) *Same 8.5 mm height and 3.2 mm pin length by JLC's attributes, gold on brass, 3 A; JLC stock 6,449 on 2026-11-11. Not chosen: the 5 mm KH-2.54FH-2X20P-H5.0 (C55160398; 12 in stock, fit on the Pi's pins unchecked)*	JLC stock (read 2026-11-11); a through-hole part on the bottom side, JLC assembly type: wave soldering	PinSocket_2x20_P2.54mm_Vertical

J3	1	**Con. 1x04**	**SM04B-SRSS-TB(LF)(SN)** JST LCSC C160404	Qwiic connector (I2C3); in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	JST_SH_SM04B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal
L1, L2	2	**SRF1260-101M** Bourns	**SRF1260-101M** Bourns LCSC C6496096	As specified; from the parts bought. JLC public stock is 92 on 2026-11-11 as well (need 40)		Nick's JLC parts library: order POB0202610210527741 (2026-10-20), qty 42, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	SRF1260-101M-Footprint-1
L3, L6	2	**PA5432.822NLT** Pulse (YAGEO group)	**PA5432.822NLT** Pulse (YAGEO group) LCSC C17230195	As specified; JLC public stock 0 on 2026-11-11, so from the parts bought		Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 42, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	FP-PA5432_472NLT-MFG
L4	1	**DFE201612E-2R2M=P2** Murata	**DFE201612E-2R2M=P2** Murata LCSC C337893	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-DFE201612C-MFG
MP1-MP4	4	**78614015360** Wurth Elektronik	**78614015360** Würth Elektronik LCSC C17256375	As specified; JLC public stock 0 on 2026-11-11, so from the parts bought		Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 84, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	78614015360-Footprint-2
R1, R44-R46	4	**RC0402FR-071K5L** Yageo / Phycomp	**RC0402FR-071K5L** Yageo LCSC C114759	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	RESC1005X40X25ML05T10
R2-R6, R9, R26, R27	8	**ERJ-2RKF4701X** Panasonic	**ERJ-2RKF4701X** Panasonic LCSC C400631	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	RESC1005X40X25LL05T05
R7, R10, R24	3	**RC0402JR-0710KL** Yageo	**RC0402JR-0710KL** Yageo LCSC C60489	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-RC0402-0_4-IPC_C
R8	1	**UR73D1JTTD10L0F** KOA Speer	**UR73D1JTTD10L0F** KOA Speer LCSC C2482319	As specified; JLC public stock 0 on 2026-11-11, so from the parts bought: 25 is 20 plus the listing's attrition of 5, no spare. Backup in JLC stock: Sofar's own listed alternate 7, Stackpole CSRF0603FT10L0 (ALT1), 9,822	CSRF0603FT10L0 (Stackpole, C346485) MFL0603R0100FA (EATON, C6090593) *Sofar's own listed alternates 7 and 1 (mote file, MANUFACTURERPARTNUMBER7 / 1); JLC stock on 2026-11-11: 9,822 and 16 (16 is short of 20). Sofar's other five: stock 0 or no listing. Not on Sofar's list, in JLC stock: ROHM PMR03EZPFU10L0 C308571 (pads not confirmed), Vishay WSLP0603R0100FEA C5334451*	Nick's JLC parts library: order DOB202610210529840 (2026-10-20), qty 25, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	RESC1608X60X55ML20T10
R11 *(DNP)*	1	**CRCW12100000Z0EA** Vishay Dale	**CRCW12100000Z0EA** Vishay Dale LCSC C844880	As specified; DNP (not fitted)		DNP: not ordered	FP-CRCW1210-e3-IPC_B
R14, R18	2	**RR0510P-101-D** Susumu	**RT0402BRD07100RL** YAGEO LCSC C705627	Specified RR0510P-101-D has JLC stock 0 (2026-11-11) and Sofar's mote file gives no alternate. 100 Ω thin film 0402, ±0.1 % ±25 ppm (tighter than the specified ±0.5 %); JLC stock 345,748 (need 48)	ERA2AEB101X (Panasonic, C445575) *100 Ω ±0.1 % ±25 ppm thin film 0402 (tighter); ALT1 AEC-Q200; PoDL front end: Sofar review*	JLC stock (read 2026-11-11); not Sofar's part, see PLANNED NOTE	FP-RR0510-IPC_B

R15, R16	2	**CRCW08057R50FKEAHP** Vishay	**CRCW08057R50FKEAHP** Vishay Dale LCSC C313962	As specified (pulse-rated part of the bus damper); Sofar's mote file gives no alternate. JLC public stock 0 on 2026-11-11, so from the parts bought	SR0805FR-477R5L (YAGEO, C873769) *Not on Sofar's list: 7.5 Ω ±1 % 0.5 W, ±200 vs ±100 ppm/°C, pulse rating not stated; JLC stock 5,492 on 2026-11-11*	Nick's JLC parts library: order DOB202610210529840 (2026-10-20), qty 50, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	CRCW08057R50FKEAHP-Footprint-1
R20, R39	2	**CRCW04021M82FKED** Vishay Dale	**RTT021824FTH** RALEC LCSC C166520	Specified CRCW04021M82FKED has JLC stock 0 (2026-11-11) and Sofar's mote file gives no alternate. Same spec: 1.82 MΩ ±1 % 62.5 mW 50 V ±100 ppm 0402; JLC stock 7,566 (need 50)		JLC stock (read 2026-11-11); not Sofar's part, see PLANNED NOTE	FP-CRCW0402-e3-MFG
R21, R35-R37, R43	5	**RMCF0402FT100K** Stackpole	**RMCF0402FT100K** Stackpole LCSC C7055151	As specified: Sofar's part is in JLC stock now (1,058 on 2026-11-11, need 110). Planned before: UNI-ROYAL 0402WGF1003TCE (JLC Basic), now ALT1	0402WGF1003TCE (UNI-ROYAL, C25741) RC0402FR-07100KL (YAGEO, C60491) *100 kΩ ±1 % 62.5 mW 50 V ±100 ppm; ALT1 is JLC Basic; AEC-Q200 not stated (Stackpole is)*	JLC stock (read 2026-11-11)	FP-RMCF0402-IPC_C
R22, R40	2	**RC0402FR-07200KL** Yageo / Phycomp	**RC0402FR-07200KL** Yageo LCSC C114763	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	RESC1005X40X25LL05T10
R23	1	**RC0402FR-0722K1L** Yageo	**RC0402FR-0722K1L** Yageo LCSC C163478	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-RC0402-0_4-IPC_B
R34	1	**RC0402FR-079K09L** Yageo / Phycomp	**RC0402FR-079K09L** Yageo LCSC C274897	As specified: in JLC stock now (4,378 on 2026-11-11, need 30). Planned before: YAGEO AC0402FR-079K09L, now ALT1	AC0402FR-079K09L (YAGEO, C227284) 0402WGF9091TCE (UNI-ROYAL, C11549) *9.09 kΩ ±1 % 62.5 mW 50 V ±100 ppm thick film; our part (D18), no amendment*	JLC stock (read 2026-11-11)	FP-RC0402-0_4-IPC_C
R38	1	**0402WGF1332TCE** UNI-ROYAL	**0402WGF1332TCE** UNI-ROYAL LCSC C122547	13.3k 1% 0402 for 5V buck at 5.11 V (was 13.7k); in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-RC0402-0_4-IPC_B
R41	1	**RC0402FR-071ML** Yageo	**RC0402FR-071ML** Yageo LCSC C138033	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	R_0402_1005Metric
R42	1	**RMCF0402FT10K0** Stackpole	**RMCF0402FT10K0** Stackpole LCSC C6111655	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-RMCF0402-IPC_C
R47	1	**RC0402FR-071KL** Yageo / Phycomp	**RC0402FR-071KL** Yageo LCSC C106235	As specified: 1 kΩ (Nick, 2026-11-11); in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	RESC1005X40X25ML05T10
T1, T2	2	**74930000** Würth Elektronik	**74930000** Würth Elektronik LCSC C4354923	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-74930000-MFG
U1	1	**ADIN2111BCPZ** Analog Devices	**ADIN2111BCPZ** Analog Devices LCSC C3226961	As specified; JLC public stock 0 on 2026-11-11, so from the parts bought. Nick's order line (Digi-Key 505-ADIN2111BCPZ-ND) carries JLC part C3226961, which JLC's public listing names ADIN2111BCPZ-R7; the listings named ADIN2111BCPZ are C3226964 and C56341494		OPEN: Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 22, still PURCHASING (JLC's estimate was 2026-11-05) on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	ADIN2111BCPZ-Footprint-1

U2, U3	2	**AP22913CN4-7** Diodes Inc.	**AP22913CN4-7** Diodes Inc. LCSC C2150198	As specified; JLC public stock 33 on 2026-11-11 (need 40), so from the parts bought		Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 42, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	AP22913CN4-7-Footprint-1
U4	1	**INA232AIDDFR** Texas Instruments	**INA232AIDDFR** Texas Instruments LCSC C5447660	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	INA232AIDDFR-Footprint-1
U5, U10	2	**LMR51430YDDCR** Texas Instruments	**LMR51430YDDCR** Texas Instruments LCSC C5210749	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	SOT-23-THIN-6_DDC_TEX
U6	1	**TPS62840YBGR** Texas Instruments	**TPS62840YBGR** Texas Instruments LCSC C2071139	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	FP-YBG0006-MFG
U11	1	**TPS26621DRCR** Texas Instruments	**TPS26621DRCR** Texas Instruments LCSC C1848341	As specified; in JLC stock (read 2026-11-11)		JLC stock (read 2026-11-11)	Texas_DRC0010J
Y1	1	**ABM11W-25.0000MHZ-6-K1Z-T3** Abracon	**ABM11W-25.0000MHZ-6-K1Z-T3** Abracon LCSC C6851222	As specified; JLC public stock 0 on 2026-11-11, so from the parts bought		Nick's JLC parts library: order DOB202610190524705 (2026-10-18), qty 22, Complete on Nick's order screen (screenshot 2026-11-11); pick it from My Parts Lib at the PCBA order	FP-ABM11W-MFG